# Hierarchical Clustering

Using on MNIST dataset

# Hierarchical Clustering Overview

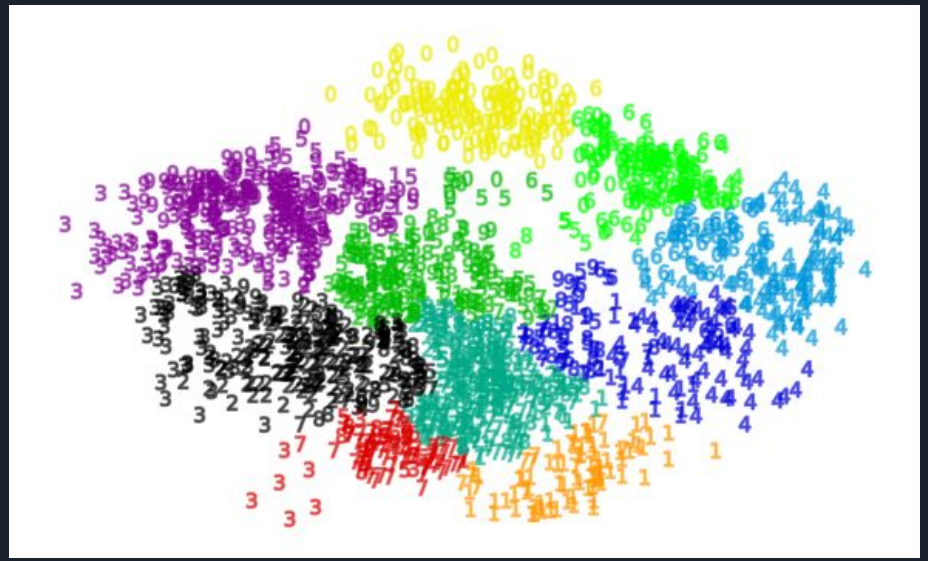
- Hierarchical clustering is a popular technique used in cluster analysis, which is a type of unsupervised machine learning
- It is used for grouping similar objects or data points into clusters based on their similarities
- Hierarchical clustering builds a hierarchy of clusters, also known as a dendrogram, where each data point is treated as a single cluster initially
- The algorithm then iteratively merges or divides these clusters until a hierarchical tree-like structure is formed.

# Bottom-Up Vs. Top-Down

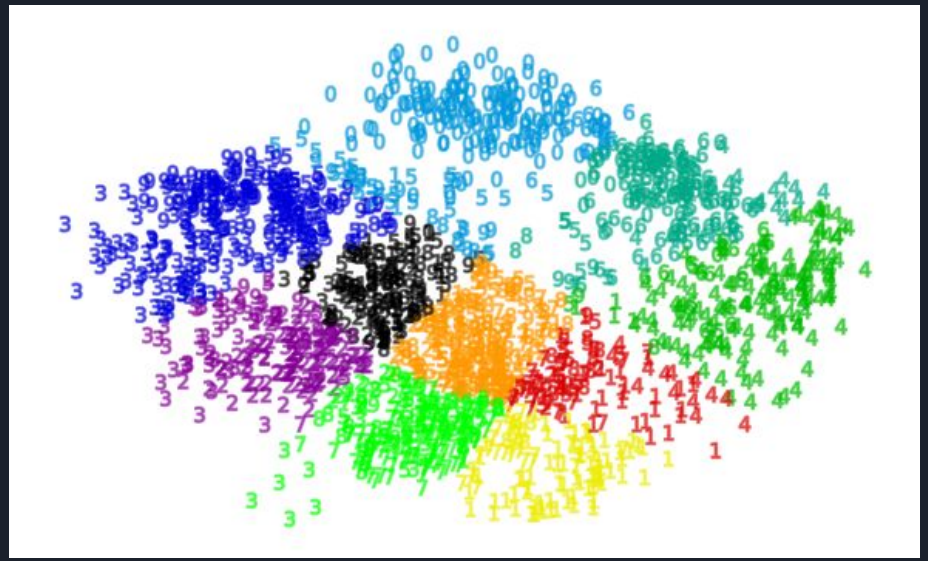
- Bottom-up approach
  - Each data point starts in its own cluster, and the algorithm recursively merges the closest pairs of clusters until a single cluster containing all the data points is obtained
  - The AgglomerativeClustering object can be used to performs a hierarchical clustering using this approach
- Top-down approach
  - All data points start in a single cluster, and the algorithm recursively splits the cluster into smaller sub-clusters until each data point is in its own cluster
  - The BisectingKMeans object can be used to performs a hierarchical clustering using this approach

# Bottom-Up Vs. Top-Down

**Bottom-Up**
PCA Components: 2
Clusters: 10



**Top-Down**
PCA Components: 2
Clusters: 10



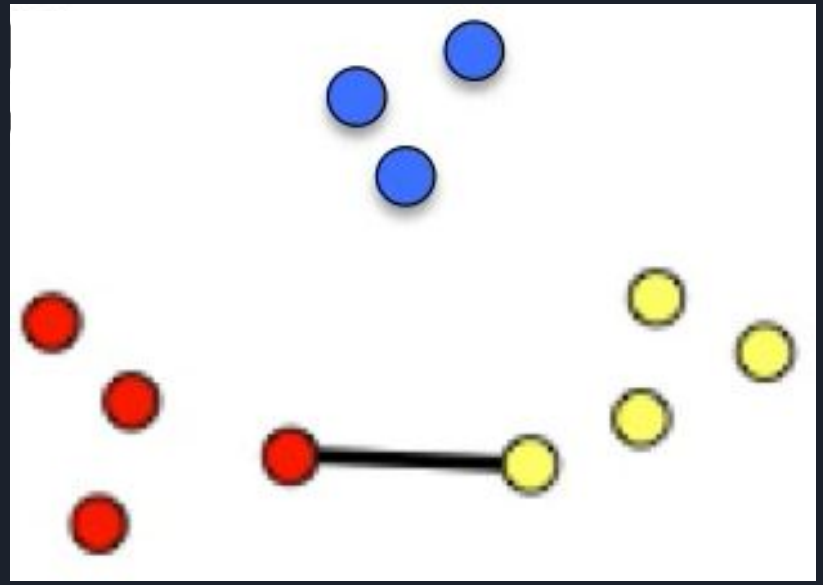
# Linkage: Single

Uses the minimum of the distances between all observations of the two sets

$$\min_{a \in A,\, b \in B} d(a, b)$$

Where:

- *A* and *B* are sets of observations
- *d* is distance



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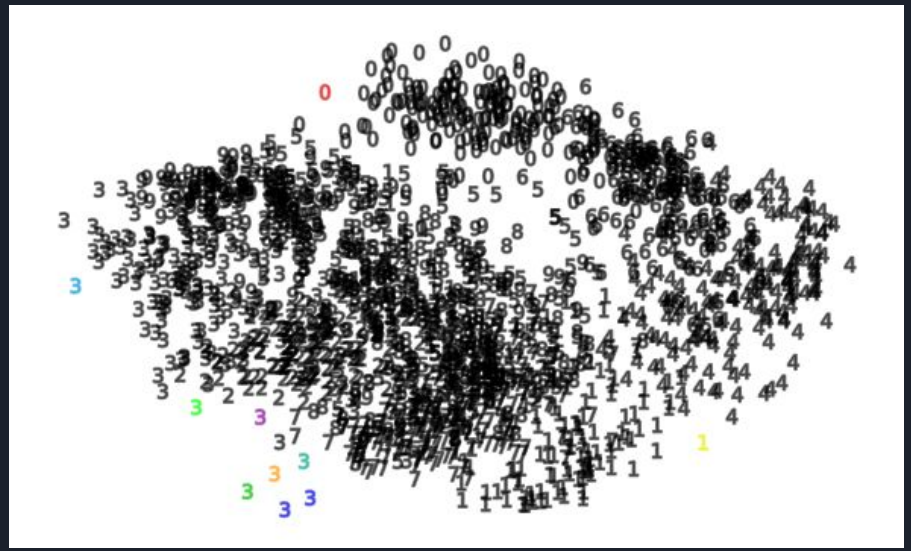
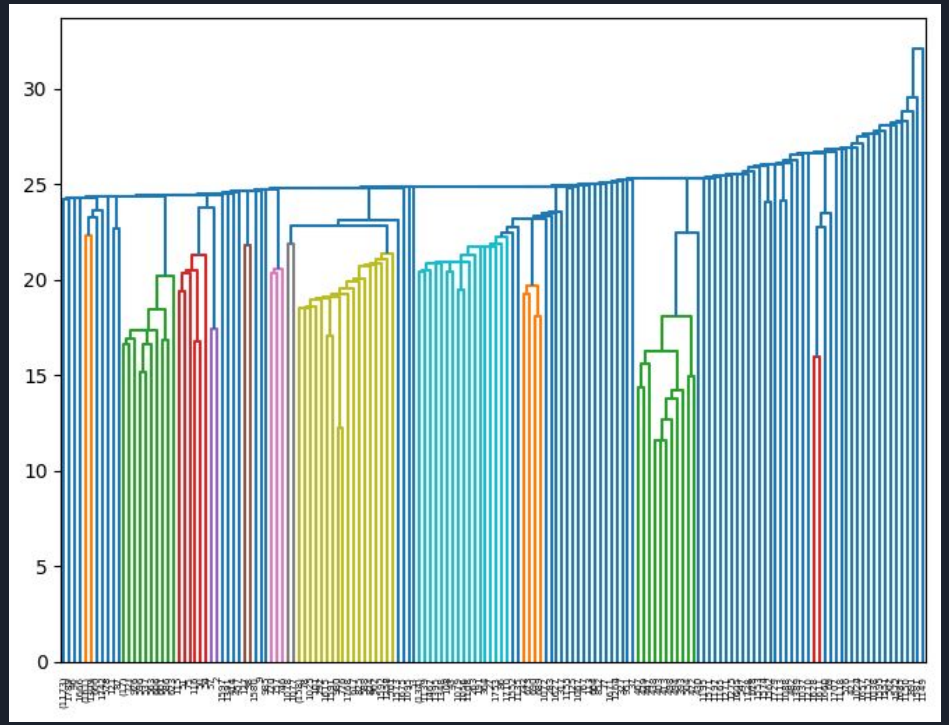
# Linkage: Single

PCA Components: 2
Clusters: 10



Truncate level: 70



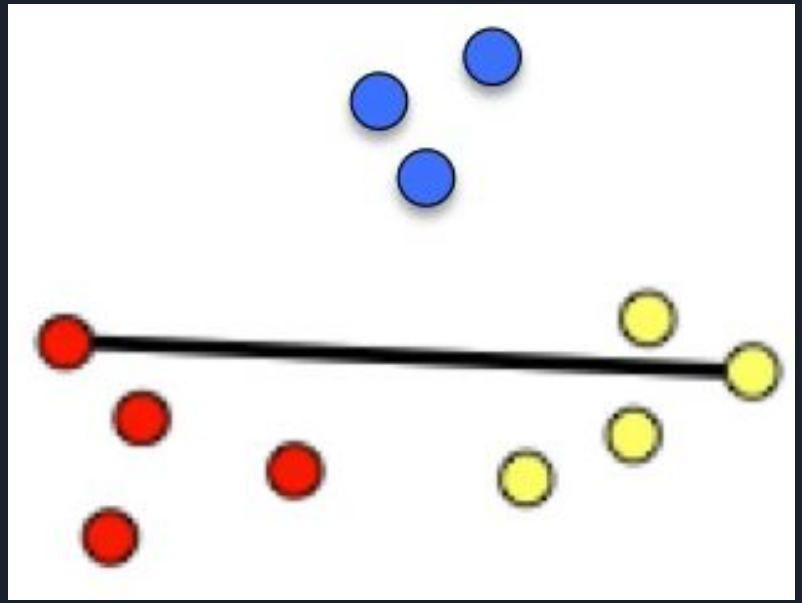
# Linkage: Complete

Uses the maximum distances between all observations of the two sets

$$\max_{a \in A,\, b \in B} d(a, b)$$

Where:

- *A* and *B* are sets of observations
- *d* is distance



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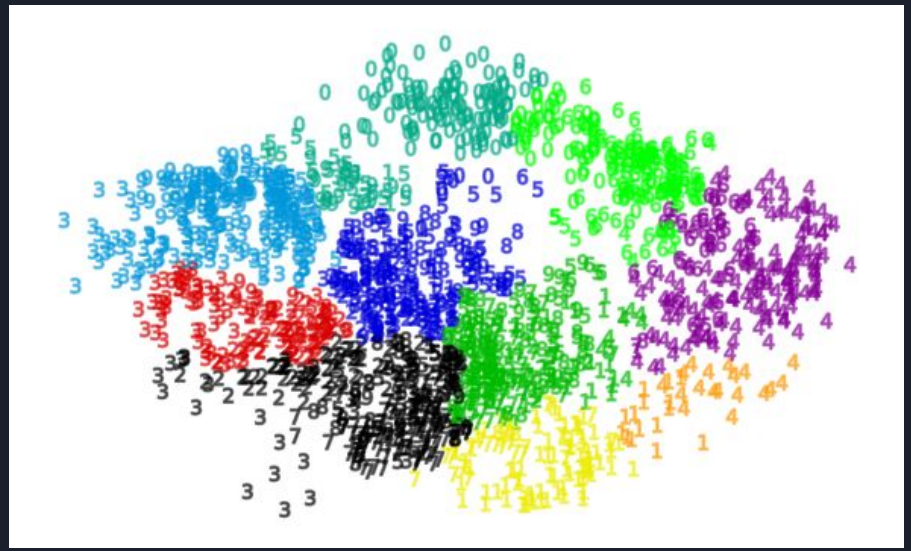
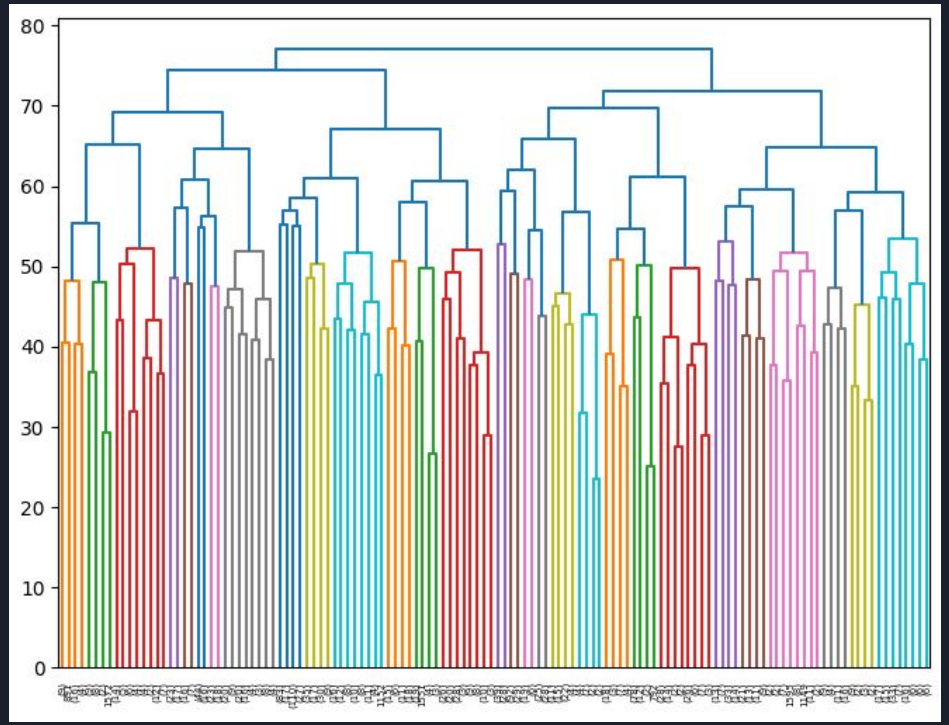
# Linkage: Complete

PCA Components: 2
Clusters: 10



Truncate level: 6



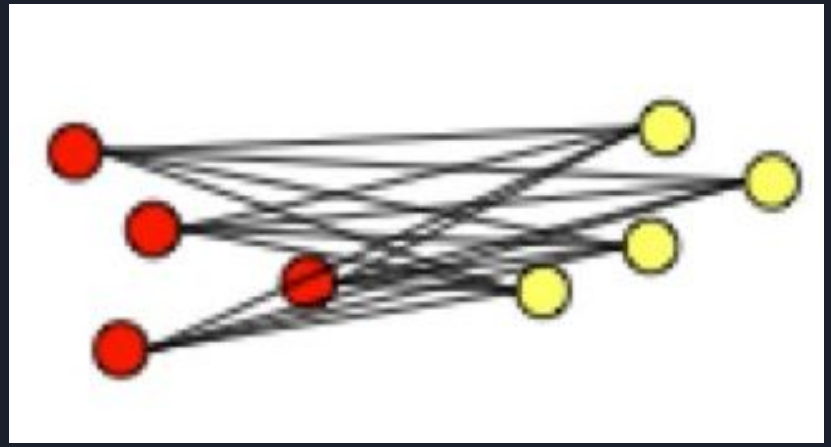
# Linkage: Average

Uses the average of the distances of each observation of the two sets

$$\frac{1}{|A| \cdot |B|} \sum_{a \in A} \sum_{b \in B} d(a, b)$$

Where:

- *A* and *B* are sets of observations
- *d* is distance



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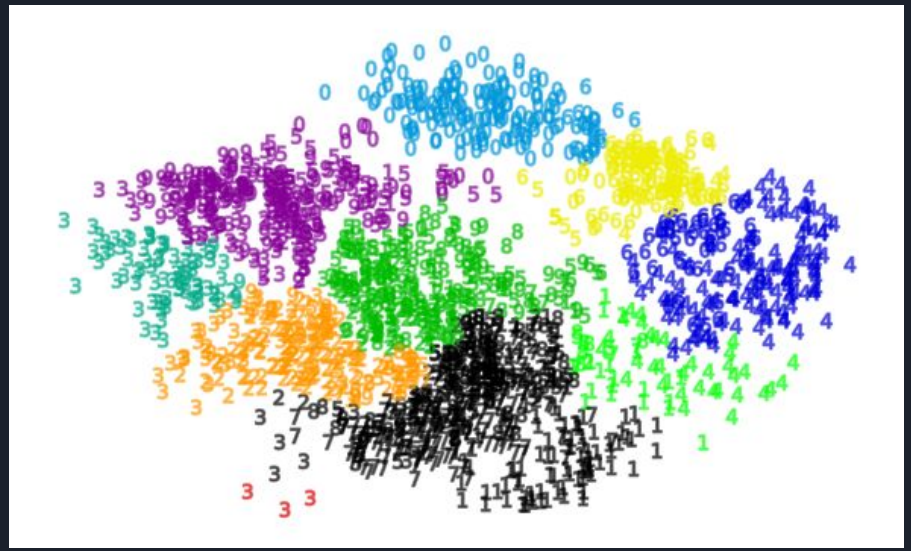
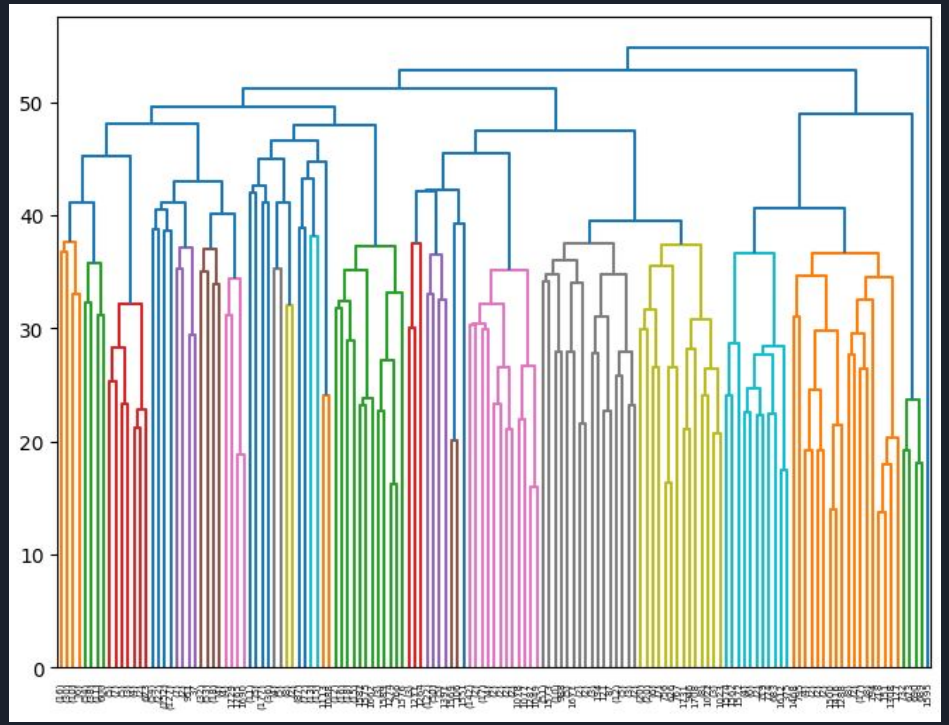
# Linkage: Average

PCA Components: 2
Clusters: 10



Truncate level: 8



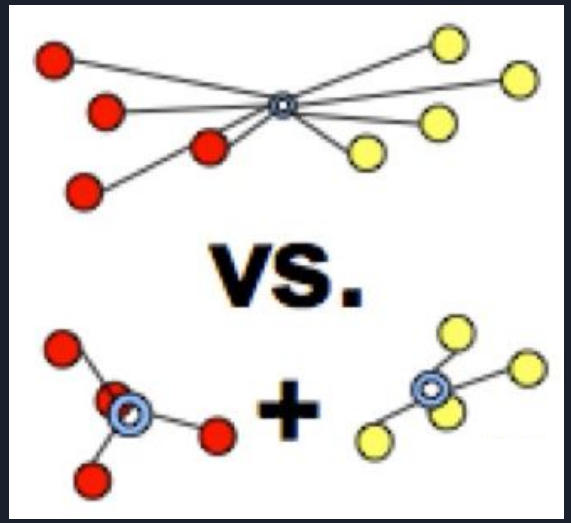
# Linkage: Ward

Minimizes the sum of squared differences within all clusters

$$\sum_{x \in A \cup B} \|x - \mu_{A \cup B}\|^2 - \sum_{x \in A} \|x - \mu_A\|^2 - \sum_{x \in B} \|x - \mu_B\|^2$$

Where:

- *A* and *B* are sets of observations
- *u_A and u_B are the centroids of A resp. B*

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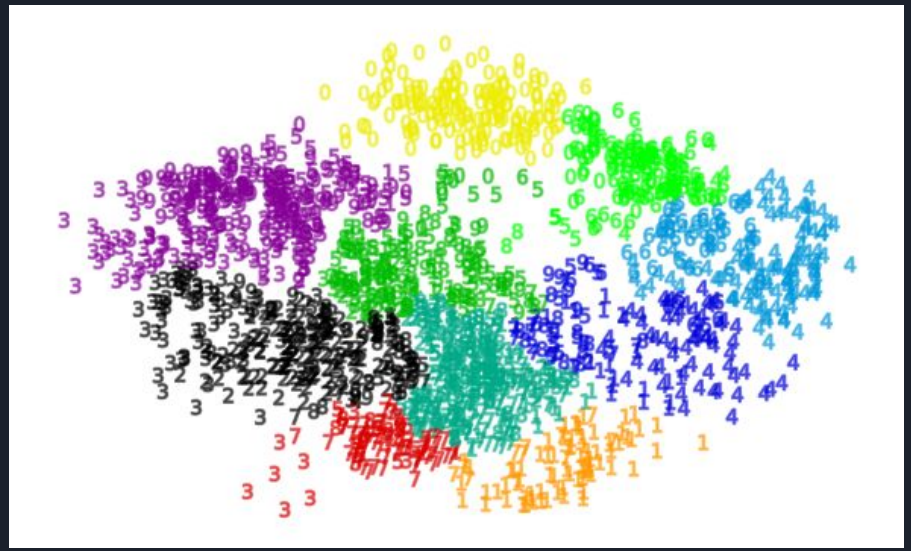
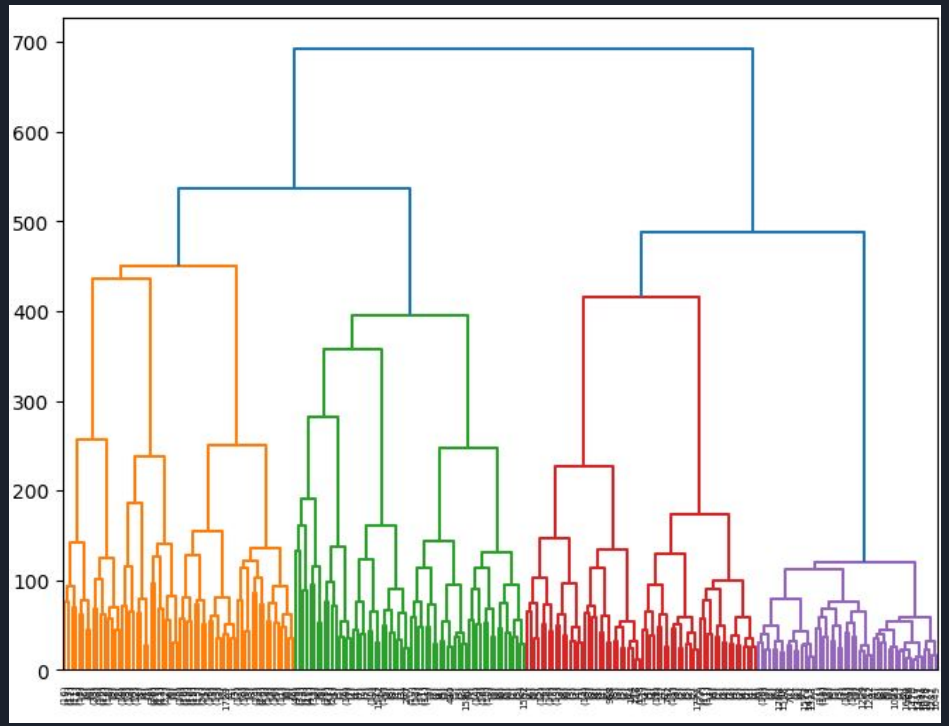
# Linkage: Ward

PCA Components: 2
Clusters: 10



Truncate level: 7



# End of Presentation